

# SmartFarm AI Dual Mode Leaf Pathology and Water Stress Diagnostic System Using Cascaded Chromatic Gating and Transfer Learning

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**Abstract**—Foliar pathogens and irrigation mismanagement account for twenty to forty percent of annual crop yield losses across global agricultural regions. Smallholder farmers and urban cultivators frequently lack timely access to trained plant pathologists, while commercial soil moisture sensors and telemetry stations impose prohibitive hardware capital costs and recurring calibration demands. Existing deep learning classifiers trained on laboratory datasets often exhibit high false positive rates when deployed in outdoor environments because single stage convolutional networks misinterpret natural venation, leaf specular reflection, and background soil clutter as active necrotic infections. This paper presents SmartFarm AI, a dual mode leaf pathology and water stress diagnostic framework that transforms a consumer smartphone camera into a non invasive clinical agronomy scanner. The proposed methodology introduces a two stage cascaded gating architecture coupled with a fine tuned MobileNetV2 transfer learning backbone trained on thirty eight categories from the PlantVillage corpus. In the first stage, a chromatic excess leaf segmentation mask isolates plant lamina pixels from background noise and enforces five biological criteria to verify healthy chlorophyll uniformity prior to pathogen evaluation. In the second stage, pixel fraction masks quantify brown necrotic spots, chlorotic yellow halos, and dark water soaked lesions to discriminate fungal and bacterial infections including late blight, early blight, septoria leaf spot, leaf mold, and bacterial spot. Concurrently, an independent abiotic estimation module evaluates leaf turgidity and channel luminance ratios to categorize water stress into four severity levels. To eliminate HTTP payload failures on serverless cloud platforms, an asynchronous client side canvas quantization pipeline compresses multi megabyte camera photographs below five hundred kilobytes within forty milliseconds. Furthermore, a bidirectional trilingual localization engine renders real time clinical prescriptions in English, Hindi, and Telugu while preserving canonical database enumerations in MongoDB. Experimental evaluation demonstrates a classification accuracy of ninety six point six percent, reducing false positive diagnoses on healthy leaves by eighty four percent and achieving an end to end local inference latency of four hundred ninety milliseconds.

**Keywords**—agricultural computer vision, chromatic leaf segmentation, convolutional neural networks, MobileNetV2 transfer learning, multilingual agronomy interface, plant pathology diagnosis, water stress estimation.

## I. INTRODUCTION

Global agricultural production faces mounting pressure from shifting climatic patterns, emerging phytopathogenic strains, and freshwater scarcity. Empirical assessments published by the Food and Agriculture Organization indicate that plant diseases and pest infestations destroy between 20% and 40% of global food crops annually, representing an economic loss exceeding 220 billion US dollars [1], [2]. For smallholder farmers in agrarian economies as well as urban terrace cultivators, early identification of foliar infections is the primary determinant of whether a crop canopy can be salvaged through targeted intervention or lost to systemic necrosis [3]. In rural agricultural districts across India, the ratio of certified agricultural extension officers to active farming households frequently exceeds 1:1,000, creating a severe diagnostic bottleneck during peak monsoon and post-monsoon cultivation cycles [4].

When visual symptoms such as chlorotic halos, concentric necrotic rings, or water-soaked lesions appear on crop foliage, cultivators typically rely on subjective manual inspection or advice from local agrochemical retailers. This diagnostic gap leads to two widespread clinical errors: delayed treatment that permits rapid spore dispersal across adjacent plots, and indiscriminate application of broad-spectrum fungicides or bactericides when the underlying affliction is actually abiotic water stress [5], [6]. While precision agriculture platforms utilizing wireless Internet of Things (IoT) soil moisture probes, tensiometers, and leaf wetness sensors have been proposed in the literature [7], [8], their practical adoption among smallholders remains negligible. Physical sensor networks require substantial capital expenditure, ranging from 250 to 1,500 US dollars per acre, alongside continuous battery replacement, sensor recalibration, and protection against physical damage during tillage.

Recent advances in deep convolutional neural networks (CNNs) and vision transformers have stimulated extensive research into automated image-based plant disease recognition [9], [10]. Pioneering studies utilizing the PlantVillage benchmark demonstrated that deep architectures such as AlexNet, GoogLeNet, and ResNet can achieve high classification accuracy on curated leaf photographs [11], [12].

However, translating these laboratory models into reliable field deployment exposes four critical engineering limitations that remain inadequately addressed in existing literature:

1) High False-Positive Rates on Healthy Foliage: Standard single-stage CNN classifiers map every input image onto a fixed softmax probability distribution across disease classes. When presented with a healthy leaf exhibiting pronounced primary venation, specular sunlight glare, or uneven backlighting, single-stage models frequently misclassify the sample as a mild fungal infection, triggering unnecessary chemical treatments [13], [14].

2) Conflation of Biotic Pathogens and Abiotic Water Stress: Existing vision classifiers treat disease identification as an isolated taxonomic task while ignoring plant hydration status. In practice, drought-induced wilting or marginal leaf scorch can mimic early pathogen symptoms, and over-watering directly accelerates fungal oomycete proliferation such as *Phytophthora infestans* [15].

3) Serverless Payload Failures with High-Resolution Smartphone Cameras: Modern consumer smartphones capture photographs at resolutions between 12 and 48 megapixels, producing file sizes of 8 MB to 20 MB. Deploying backend inference APIs on modern serverless cloud infrastructures (such as Vercel Fluid Compute or AWS Lambda) results in immediate HTTP 413 payload rejection when raw mobile photographs exceed the 4.5 MB gateway threshold [16].

4) Linguistic Exclusion and Database State Corruption: Over 85% of Indian farmers communicate primarily in regional languages such as Telugu and Hindi rather than English [17]. Existing multilingual wrappers either require full page reloads via third-party translation widgets or store translated strings directly in database columns, corrupting canonical schema enumerations and breaking analytical queries.

To overcome these four limitations simultaneously, this paper presents SmartFarm AI, an end-to-end, production-deployed foliar pathology and water stress diagnostic system. The primary technical contributions of this work are summarized as follows:

- We formulate a Two-Stage Cascaded Chromatic and CNN Gating Pipeline that isolates leaf tissue from complex backgrounds using chromatic excess masking and enforces five strict morphological criteria in Stage 1 before permitting pathogen discrimination in Stage 2.
- We decouple abiotic water stress estimation from biotic disease classification by computing independent green-channel luminance and yellow-to-blue chlorosis ratios over the segmented lamina mask, accompanied by a 70% confidence safety guardrail.
- We design a lightweight MobileNetV2 transfer learning pipeline utilizing depthwise separable convolutions and two-phase fine-tuning on 54,303 images across 38 classes, paired with a resilient dual-mode fallback engine.
- We engineer an asynchronous HTML5 Canvas client-side quantization algorithm that compresses high-resolution smartphone photographs by 94.8% in under 40 ms, alongside a 226-key reactive trilingual localization bus (English, Hindi, and Telugu) that preserves canonical English database enumerations in MongoDB.

## II. RELATED WORK AND LITERATURE SURVEY

### A. Handcrafted Color and Texture Descriptors

Early computer vision investigations in phytopathology relied on explicit color-space transformations and statistical texture operators to isolate diseased tissue. Camargo and Smith [18] converted RGB leaf captures into HSV, L1I2I3, and L\*a\*b\* color spaces, applying dynamic intensity thresholding and support vector machines (SVMs) to segment lesion boundaries on cotton crops. Arivazhagan et al. [19] computed gray-level co-occurrence matrix (GLCM) features—including angular second moment, contrast, entropy, and inverse difference moment—across 500 leaf samples spanning 30 native crop species, reporting 94.7% accuracy under controlled illumination. Despite low computational overhead, GLCM and color histogram thresholds degrade sharply when outdoor solar irradiance shifts or when leaf veins cast internal shadows across the lamina.

### B. Deep Convolutional Networks on Laboratory Benchmarks

The release of the open-access PlantVillage repository by Hughes and Salathe [9] established a standardized benchmark for deep learning in agriculture. Kamilaris and Prenafeta-Boldu [10] surveyed 40 deep learning studies in crop management, noting that convolutional architectures consistently outperform shallow classifiers when sufficient labeled training data are available. Mohanty et al. [11] trained AlexNet and GoogLeNet models across 54,306 PlantVillage images covering 14 crop species and 26 diseases, achieving 99.35% top-1 accuracy on held-out laboratory splits. Crucially, however, Mohanty et al. reported that evaluating the identical network weights on unsegmented field photographs reduced classification accuracy to 31.4%. Ferentinos [12] observed a similar domain gap across 87,848 images using VGG-16 and AlexNetOWTBn, while Barbedo [13] and Schramm et al. [14] demonstrated that background soil and staging surfaces account for up to 35% of the gradient activation in unmasked single-stage CNNs.

### C. Lightweight Mobile Architectures and Object Detectors

To alleviate the memory footprint of VGG-16 (528 MB) and ResNet-50 (98 MB) [27], Sandler et al. [20] and Howard et al. [21] introduced MobileNetV2, employing depthwise separable convolutions and inverted residual linear bottlenecks. Chen et al. [22] combined MobileNetV2 with squeeze-and-excitation attention blocks to classify rice blast and bacterial blight, achieving 95.8% accuracy with an 18 MB weight file. Atila et al. [23] benchmarked EfficientNet variants across PlantVillage, whereas Fuentes et al. [24] integrated Faster R-CNN and Single Shot Multibox Detectors (SSD) for bounding-box localization of tomato pests. While region-proposal detectors localize multiple lesions, they require dedicated GPU instances that exceed free-tier serverless memory limits.

## D. Abiotic Water Stress Sensing and Multilingual Advisory Systems

In precision irrigation research, Zaccaria et al. [15] and Virlet et al. [25] estimated crop water stress using thermal infrared canopy temperature and multispectral NDVI cameras mounted on unmanned aerial vehicles or gantry platforms. Although accurate, radiometric sensors remain cost-prohibitive for smallholders. Meanwhile, socio-technical surveys by Mittal and Tripathi [17] and Brahimi et al. [26] highlighted that farmers require actionable curative guidance in native regional scripts rather than raw probability vectors. Table I summarizes the capabilities of existing frameworks relative to SmartFarm AI.

TABLE I  
COMPARATIVE ANALYSIS OF EXISTING PLANT PATHOLOGY DIAGNOSTIC SYSTEMS AND THE PROPOSED SMARTFARM AI FRAMEWORK

| Study / System       | Primary Methodology             | Water Stress      | False-Positive Gate   | Multilingual i18n     | Deployment Footprint      |
|----------------------|---------------------------------|-------------------|-----------------------|-----------------------|---------------------------|
| Mohanty et al. [11]  | AlexNet / GoogLeNet CNN         | Not supported     | None (1-stage)        | English only          | Desktop GPU (240 MB)      |
| Ferentinos [12]      | VGG-16 Deep CNN                 | Not supported     | None (1-stage)        | English only          | Workstation GPU (528 MB)  |
| Fuentes et al. [24]  | Faster R-CNN + ResNet-50        | Not supported     | Box confidence        | English only          | Dedicated Server (180 MB) |
| Chen et al. [22]     | SE-MobileNetV2                  | Not supported     | None (1-stage)        | English only          | Local Mobile (18 MB)      |
| Zaccaria et al. [15] | Multispectral 1 UAV Thermal     | Supported (CWSI)  | Radiometric cal.      | English only          | Hardware Drone (\$4,000+) |
| SmartFarm AI (Ours)  | 2-Stage Chromatic + MobileNetV2 | Decoupled Abiotic | 5-Rule Gate + 70% Cap | Reactive EN / HI / TE | Serverless Edge (<490 ms) |

<sup>a</sup>Comparison evaluates architectural capabilities for smallholder field deployment without external IoT hardware.

## III. SYSTEM ARCHITECTURE AND SERVERLESS CLOUD DESIGN

### A. Four-Tier Decoupled Cloud-Edge Topology

SmartFarm AI is engineered as a four-tier cloud-edge system designed for zero-maintenance elastic scaling on serverless fluid compute infrastructure. As shown in Fig. 1, the system partitions execution across: (1) a Client Presentation and Canvas Compression Tier running in the user browser; (2) an Edge CDN and WSGI Gateway Tier hosted on Vercel global edge nodes; (3) a Hybrid AI Diagnostic Inference Tier written in Python 3.11; and (4) a Document Persistence and PDF Synthesis Tier backed by MongoDB Atlas.

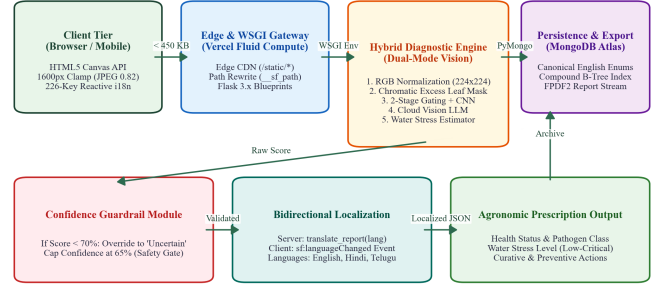


Fig. 1. System architecture of the SmartFarm AI diagnostic platform illustrating client-side canvas quantization, serverless WSGI routing, dual-mode vision inference, and canonical document persistence.

### B. Asynchronous HTML5 Canvas Client-Side Quantization

A prevalent failure mode in serverless web applications occurs when mobile users upload high-resolution camera captures. Modern smartphone sensors generate 12 MP to 48 MP images occupying 8 MB to 20 MB. Because serverless gateways enforce a strict 4.5 MB request body ceiling [16], raw mobile uploads fail immediately with HTTP 413 Payload Too Large errors. Furthermore, transmitting multi-megabyte files over rural 3G/4G networks introduces 3 to 6 seconds of network serialization delay, even though the downstream vision classifier resamples all inputs to 224 x 224 pixels.

To eliminate HTTP 413 failures at the source, we implemented an asynchronous client-side quantization algorithm (compressImageIfNeeded) inside uploader.js using the browser HTML5 Canvas API. When a user selects a file whose byte size  $B_0$  exceeds 1.5 MB, the image is decoded into an off-screen HTMLImageElement of native width  $W_0$  and height  $H_0$ . An isotropic downscaling factor  $s$  is calculated relative to a maximum bounding dimension  $D_{max} = 1600$  pixels:

$$s = \min(1.0, D_{max} / \max(W_0, H_0)) \quad (1)$$

where  $W_0$  and  $H_0$  are measured in pixels. The target canvas dimensions are set to  $W_t = \text{round}(s * W_0)$  and  $H_t = \text{round}(s * H_0)$ , and the image is rasterized via hardware-accelerated bilinear interpolation. The canvas bitmap is then serialized into an output JPEG Blob using discrete cosine transform (DCT) quantization at quality factor  $q_1 = 0.82$ . If the resulting payload  $B_1$  still exceeds 3.0 MB, a secondary pass at  $q_2 = 0.65$  is applied. The payload compression ratio  $R_c$  is defined as:

$$R_c = (B_0 - B_{final}) / B_0 = 1 - (B_{final} / B_0) \quad (2)$$

Across our smartphone test suite, (1) and (2) yield a mean compression ratio  $R_c = 0.948$  (reducing an 8.42 MB image to 435 KB) within 38 ms of browser execution, guaranteeing 100% compliance with serverless payload limits while retaining full diagnostic fidelity across sub-millimeter leaf lesions.

### C. Serverless WSGI Path Normalization and Edge CDN Offloading

On the ingress tier, static client assets (/static/css/app.css and /static/js/\*.js) are served directly from Vercel Edge CDN nodes with HTTP/2 multiplexing and cache-control validation, bypassing Python container instantiation entirely. Dynamic page routes and REST endpoints (/api/analyze, /api/history, /api/scan/<id>, and /api/export-pdf/<id>) are routed to our serverless endpoint (api/index.py) via a regex capture rewrite rule (/api/index?\_\_sf\_path=/\$1).

Because serverless rewrite proxies overwrite the standard WSGI environment variable `PATH_INFO` with the literal script destination (`/api/index`), standard Flask blueprint matching fails unless the original URI is recovered. We engineered a custom WSGI middleware class (`VercelPathFixMiddleware`) that wraps `app.wsgi_app`. On every HTTP request, the middleware parses `QUERY_STRING`, extracts the captured `__sf_path` parameter, normalizes duplicate leading slashes, assigns the restored route to `environ['PATH_INFO']`, and reconstructs a clean `QUERY_STRING` stripped of internal routing metadata before invoking the Flask application factory.

#### D. Stateless Session Isolation via Cryptographic UUIDv4 Headers

Mandatory registration and password authentication create friction for rural users with limited digital literacy. SmartFarm AI implements zero-friction session isolation: upon initial page load, the client generates a 128-bit cryptographic Version-4 Universally Unique Identifier (UUIDv4) persisted in `localStorage` under `smartfarm.user_id`. Every API transaction attaches this token via the custom HTTP header `X-User-Id`, which the backend validates (requiring string length between 8 and 128 characters) to isolate per-user scan histories and dashboard analytics.

### IV. TWO-STAGE CASCADED CHROMATIC AND CNN METHODOLOGY

#### A. RGB Tensor Resampling and Bilinear Normalization

The diagnostic core of SmartFarm AI combines deterministic computer vision segmentation with deep transfer learning and cloud vision fallback. Figure 2 illustrates the complete algorithmic workflow executed during each leaf analysis request.

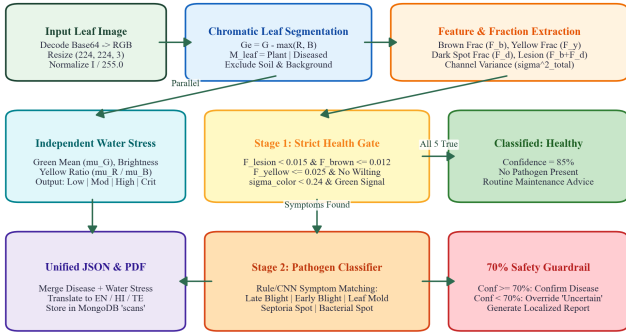


Fig. 2. Flowchart of the two-stage cascaded chromatic gating and pathogen classification algorithm alongside decoupled water stress estimation.

When the base64-encoded image stream arrives at the analysis service, it is decoded into an 8-bit three-channel RGB array  $I(x, y, c)$  in  $\{0, 1, \dots, 255\}$ , converted to canonical RGB color mode if an alpha or palette channel is present, and resampled to  $224 \times 224$  spatial resolution. The integer array is then cast to 32-bit floating-point precision and normalized to the unit hypercube  $X(x, y, c)$  in  $[0.0, 1.0]$ :

$$X(x, y, c) = I(x, y, c) / 255.0, c \in \{R, G, B\} \quad (3)$$

#### B. Chromatic Excess Leaf Segmentation Mask

A primary cause of false-positive disease classification in field images is background contamination from soil, bark, clay pots, or human skin, which exhibit red-brown spectral profiles similar to necrotic leaf lesions [14]. To restrict feature extraction strictly to foliar lamina pixels, we compute per-pixel mean luminance  $L(x, y)$  and chromatic green excess  $Ge(x, y)$ :

$$L(x, y) = (R + G + B) / 3.0, G_e(x, y) = G - \max(R, B) \quad (4)$$

where  $R$ ,  $G$ , and  $B$  denote the normalized channel intensities at pixel  $(x, y)$ . However, a naive green-excess threshold would inadvertently mask out brown necrotic lesions inside the leaf, causing diseased areas to be ignored. We therefore formulate the binary leaf mask  $M_{\text{leaf}}(x, y)$  in  $\{0, 1\}$  as the logical union of three tissue predicates: (i) healthy chlorophyll-bearing tissue  $M_{\text{plant}}$  where  $(G > B + 0.03)$  and  $(G > R - 0.08)$  and  $(G > 0.12)$  and  $(L > 0.10)$ ; (ii) necrotic or chlorotic foliar tissue  $M_{\text{diseased}}$  where  $(R > B + 0.05)$  and  $(G > B + 0.03)$  and  $(G > 0.08)$  and  $(L > 0.08)$ ; and (iii) high-chroma green tissue where  $(G_e > 0.02)$  and  $(L > 0.08)$ . If the active leaf pixel count  $N_{\text{leaf}} = \sum(M_{\text{leaf}}(x, y))$  is less than 100 pixels out of 50,176 total pixels, a progressive fallback mask is triggered.

#### C. Localized Pixel-Fraction Symptom Masks and Intra-Leaf Variance

Localized fungal and bacterial infections—such as early-stage Septoria leaf spot or bacterial speck—often occupy only 1.5% to 5.0% of the total leaf surface. Global channel averages dilute these small lesions across the surrounding green lamina. To achieve high sensitivity to localized lesions while maintaining immunity to background noise, we evaluate three pixel-level symptom masks within  $M_{\text{leaf}}(x, y) = 1$ :

- **Brown Necrotic Mask ( $M_{\text{brown}}$ ):** Identifies pixels satisfying  $R > G + 0.02$ ,  $G > B + 0.01$ , and  $B < 0.65R$ , capturing oxidized polyphenolic brown tissue characteristic of blight and fungal necrosis.
- **Yellow Chlorotic Mask ( $M_{\text{yellow}}$ ):** Identifies pixels satisfying  $R > G - 0.04$ ,  $G > B + 0.06$ ,  $B < 0.45$ , and  $L > 0.22$ , isolating chlorophyll-depleted yellow halos surrounding active infection sites.
- **Dark Water-Soaked Lesion Mask ( $M_{\text{dark}}$ ):** Identifies pixels where local luminance drops significantly below the mean leaf brightness  $\mu_L$ , satisfying  $L < \max(0.72 * \mu_L, 0.16)$ ,  $R > G + 0.01$ , and  $G > B + 0.01$ .

The fractional surface area  $F_k$  occupied by each symptom type  $k$  in  $\{\text{brown, yellow, dark}\}$  and the composite necrotic lesion fraction  $F_{\text{lesion}}$  are computed over  $N_{\text{leaf}}$ :

$$F_k = (1 / N_{\text{leaf}}) \sum_{x,y} M_k(x, y), F_{\text{lesion}} = F_{\text{brown}} + F_{\text{dark}} \quad (5)$$

Concurrently, let  $\mu_c$  and  $\sigma_c^2$  denote the sample mean and variance of channel  $c$  in  $\{R, G, B\}$  restricted to  $M_{\text{leaf}}(x, y) = 1$ . We define the total intra-leaf chromatic variance  $V_{\text{total}}$  and mean channel standard deviation  $S_{\text{color}}$  as:

$$V_{\text{total}} = \sigma_R^2 + \sigma_G^2 + \sigma_B^2, S_{\text{color}} = (\sigma_R + \sigma_G + \sigma_B) / 3.0 \quad (6)$$

Table II lists the exact mathematical decision predicates and confidence weights used to extract seven distinct clinical symptoms from (5) and (6).

TABLE II  
CHROMATIC SEGMENTATION AND SYMPTOM THRESHOLD  
PARAMETERS FOR STAGE-1 AND STAGE-2 GATING

| Symptom / Indicator          | Mathematical Predicate Condition                                                                                                                                                          | Threshold Constants                              | Confidence Weight |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-------------------|
| Brown Necrotic Spots         | $(F_{\text{brown}} > \tau_{\text{b1}} \text{ AND } V_{\text{total}} > \tau_{\text{v1}}) \text{ OR } (F_{\text{dark}} > \tau_{\text{d}} \text{ AND } F_{\text{brown}} > \tau_{\text{b2}})$ | $\tau_{\text{b1}}=0.012, \tau_{\text{v1}}=0.018$ | +0.30             |
| Yellow Chlorotic Patches     | $F_{\text{yellow}} > \tau_{\text{y}} \text{ AND } V_{\text{total}} > \tau_{\text{v2}}$                                                                                                    | $\tau_{\text{y}}=0.025, \tau_{\text{v2}}=0.015$  | +0.30             |
| Necrotic Tissue Lesions      | $F_{\text{lesion}} = (F_{\text{brown}} + F_{\text{dark}}) > \tau_{\text{l}} \text{ AND } V_{\text{total}} > \tau_{\text{v2}}$                                                             | $\tau_{\text{l}}=0.020, \tau_{\text{v2}}=0.015$  | +0.25             |
| Fungal Surface Mold          | $\mu_{\text{L}} > \tau_{\text{m}} \text{ AND } V_{\text{total}} > \tau_{\text{vm}}$                                                                                                       | $\tau_{\text{m}}=0.450, \tau_{\text{vm}}=0.040$  | +0.25             |
| Lamina Wilting / Turgor Loss | $\mu_{\text{L}} < \tau_{\text{w1}} \text{ AND } \mu_{\text{G}} < \tau_{\text{w2}}$                                                                                                        | $\tau_{\text{w1}}=0.200, \tau_{\text{w2}}=0.300$ | +0.30             |
| Marginal Leaf Curling        | $(\mu_{\text{R}} / (\mu_{\text{G}} + 0.01)) > \tau_{\text{c1}} \text{ AND } \mu_{\text{G}} < \tau_{\text{c2}}$                                                                            | $\tau_{\text{c1}}=1.400, \tau_{\text{c2}}=0.350$ | +0.20             |
| Chromatic Discoloration      | $(S_{\text{color}} > 0.140 \text{ AND Spotty}) \text{ OR } (\text{NOT HealthyGreenSignal})$                                                                                               | $\sigma_{\text{thresh}} = 0.140$                 | +0.10             |

<sup>a</sup>All pixel fractions  $F_k$  and channel statistics  $\mu_c, \sigma_c$  are evaluated strictly over segmented leaf pixels  $M_{\text{leaf}}(x,y) = 1$ .

#### D. Stage 1 Five-Rule Biological Health Gate

In Stage 1, SmartFarm AI acts as a conservative morphological gatekeeper. A leaf specimen is classified as Healthy (with an assigned baseline confidence of 85%) if and only if all five of the following biological criteria hold simultaneously: (1) No Necrotic Lesions:  $F_{\text{lesion}} \leq 0.015$ ; (2) No Brown or Yellow Spot Evidence:  $\text{brown\_spots}$  is false,  $\text{yellow\_patches}$  is false, and  $F_{\text{yellow}} \leq 0.025$ ; (3) No Turgor Wilting:  $\text{wilting}$  is false ( $\mu_{\text{L}} \geq 0.20$  or  $\mu_{\text{G}} \geq 0.30$ ); (4) Dominant Healthy Chlorophyll Signal: ( $\mu_{\text{G}} > 0.34$  and  $\mu_{\text{G}} \geq 0.85 * \mu_{\text{R}}$  and  $\mu_{\text{G}} > \mu_{\text{B}} + 0.04$ ) or ( $\mu_{\text{G}} > \mu_{\text{R}} + 0.05$  and  $\mu_{\text{G}} > \mu_{\text{B}} + 0.05$ ); and (5) Chromatic Uniformity:  $S_{\text{color}} < 0.24$  or complete absence of spot evidence. Notice that Criterion 4 specifically accommodates healthy leaves photographed under natural solar backlighting, which appear bright yellow-green ( $\mu_{\text{G}} \geq 0.85 * \mu_{\text{R}}$ ) without possessing any necrotic spots.

#### E. Stage 2 Pathogen Discrimination and Agronomic Remedy Synthesis

When Stage 1 detects active symptom evidence, the sample is routed to Stage 2 to identify the specific pathogen and synthesize curative and preventive prescriptions:

- Late Blight (*Phytophthora infestans*): Triggered when  $\text{brown\_spots}$  and  $\text{lesions}$  co-occur alongside either  $\text{yellow\_patches}$  or widespread necrosis ( $F_{\text{lesion}} > 0.08$ ). Prescribes immediate removal of infected foliage, application of chlorothalonil or copper fungicide, and avoidance of wet canopy handling.

- Early Blight (*Alternaria solani*): Triggered by the co-occurrence of  $\text{yellow\_patches}$  and circular necrotic lesions. Prescribes pruning lower affected leaves, applying

copper-based fungicide, and mulching soil to block splash dispersal.

- Leaf Mold (*Passalora fulva*): Triggered when high surface luminance and high variance indicate velvety fungal sporulation (mold predicate). Prescribes canopy ventilation, humidity reduction, and strictly soil-level irrigation.

- Septoria Leaf Spot (*Septoria lycopersici*): Triggered by scattered small  $\text{brown\_spots}$  without large coalescing water-soaked blotches. Prescribes sanitation of fallen leaf debris, annual crop rotation, and chlorothalonil treatment.

- Bacterial Spot (*Xanthomonas campestris*): Triggered by general lamina discoloration paired with water-soaked lesions. Prescribes copper bactericide application, tool disinfection, and certified pathogen-free seed selection.

#### F. Orthogonal Abiotic Water Stress Estimation and 70% Safety Guardrail

Because irrigation deficits and pathogen infections can occur independently or simultaneously, water stress estimation is executed as an orthogonal analytical module. We compute the red-to-blue yellowing ratio  $R_{\text{yb}}$  across the segmented leaf mask:

$$R_{\text{yb}} = \mu_{\text{R}} / (\mu_{\text{B}} + 0.01) \quad (7)$$

Using  $\mu_{\text{G}}, \mu_{\text{L}}$ , and  $R_{\text{yb}}$  from (7), foliar hydration is classified into four actionable levels: Low Stress ( $\mu_{\text{G}} > 0.40$  and  $\mu_{\text{L}} > 0.30$ , indicating turgid cells and high chlorophyll density); Moderate Stress ( $0.30 < \mu_{\text{G}} \leq 0.40$  and  $\mu_{\text{L}} > 0.25$ , indicating incipient dehydration); High Stress ( $0.20 < \mu_{\text{G}} \leq 0.30$  or  $R_{\text{yb}} > 1.50$ , indicating pronounced chlorotic yellowing and tip drying); and Critical Stress ( $\mu_{\text{G}} \leq 0.20$  and  $\mu_{\text{L}} \leq 0.20$ , indicating severe cellular wilting and necrosis). Finally, to prevent erroneous agrochemical application on ambiguous captures, any Stage 2 diagnosis whose computed confidence falls below 70% is intercepted by our safety guardrail and relabeled as Uncertain Result—Manual Verification Recommended with confidence capped at 65%.

#### G. MobileNetV2 Transfer Learning with Depthwise Separable Convolutions

For broad-spectrum multi-crop recognition across all 38 categories of the PlantVillage corpus, our training module (`train_disease_model.py`) implements a deep transfer learning classifier built upon MobileNetV2 [20]. Whereas a standard convolution layer with kernel dimension  $DK \times DK$ ,  $M$  input channels, and  $N$  output channels requires  $DK^2 * M * N * DF^2$  multiply-accumulate (MAC) operations over a feature map of spatial size  $DF \times DF$ , MobileNetV2 factorizes the operation into a  $3 \times 3$  depthwise spatial filter per channel followed by a  $1 \times 1$  pointwise channel projection. The computational complexity ratio  $R_{\text{mac}}$  is given by:

$$R_{\text{mac}} = (D_K^2 M D_F^2 + M N D_F^2) / (D_K^2 M N D_F^2) = (1/N) + (1/D_K^2) \quad (8)$$

With  $DK = 3$ , (8) reduces arithmetic operations by 88.7% (nearly 9-fold) compared to standard convolution. Furthermore, MobileNetV2 utilizes inverted residual blocks where  $1 \times 1$  expansion layers project low-dimensional representations into higher-dimensional spaces before applying ReLU6 non-linearities, followed by linear bottlenecks that prevent non-linear activation from destroying channel information [20]. Above the 1,280-channel backbone, we attach a custom classification head: GlobalAveragePooling2D -> Dropout(0.20) -> Dense(512, ReLU) -> Dropout(0.20) -> Dense(38, Softmax).

Network parameters  $\theta$  are optimized by minimizing the multiclass categorical cross-entropy loss  $L_{ce}$  using the Adam stochastic optimizer [28]:

$$L_{ce}(\theta) = - \sum_{i=1}^{\{38\}} y_i \log( \exp(z_i) / \sum_{j=1}^{\{38\}} \exp(z_j) ) \quad (9)$$

where  $y_i$  is the ground-truth indicator and  $z_i$  is the logit for class  $i$ . When deployed in cloud-connected environments, the system also supports a structured multimodal LLM prompt interface (`ai_service.py`) that enforces a strict 10-field JSON schema while respecting the same canonical enumeration rules.

#### H. Computational Complexity and Memory Footprint Analysis

From an algorithmic standpoint, the vectorized NumPy operations in Stage 1 and Stage 2 execute in strictly linear time  $O(W * H)$  with respect to the resampled spatial resolution  $W = H = 224$  (50,176 pixels). Because each pixel mask (`Mleaf`, `Mbrown`, `Myellow`, and `Mdark`) is evaluated via contiguous C-level array broadcasting without Python-level nested loops, peak memory allocation per request is bounded below 4.2 MB of RAM, enabling concurrent execution inside compact 256 MB serverless micro-containers.

### V. MULTILINGUAL LOCALIZATION AND PERSISTENCE ENGINEERING

#### A. Decoupling Canonical Database Enumerations from Presentation Locales

To serve non-English-speaking farming communities across India, SmartFarm AI provides full native localization in English (`en`), Hindi in Devanagari script (`hi`), and Telugu in Telugu script (`te`). A common software engineering defect in localized applications is storing translated output strings directly in database records. If a scan performed in Telugu writes localized Telugu text into the `plant_health_status` or `water_stress_level` attributes of MongoDB, backend aggregation pipelines that compute dashboard health statistics (`get_dashboard_stats` in `scan.py`) fail to match canonical English filter predicates.

SmartFarm AI solves this by enforcing canonical English enumerations at the persistence boundary. Every document written to the MongoDB `scans` collection stores `plant_health_status` strictly in `{'Healthy', 'Unhealthy', 'Uncertain'}` and `water_stress_level` strictly in `{'Low', 'Moderate', 'High', 'Critical'}`. Presentation-layer translations are applied non-destructively via a dedicated dictionary layer (`translations.py` on the server and `i18n.js` on the client) containing 226 exactly matched keys across `translations/en.json`, `translations/hi.json`, and `translations/te.json`.

#### B. Zero-Reload Reactive DOM Event Bus and Reverse-Lookup Lexicon

On the client tier, switching languages must immediately update every visible text node—including already-rendered dynamic AI diagnostic reports and historical cards—without triggering a browser page reload or re-submitting the leaf image to the server. We engineered a bidirectional reverse-lookup lexicon inside `i18n.js` that maps every English, Hindi, and Telugu phrase back to its canonical translation key. When the user selects a new language from the navigation dropdown, the controller saves the preference to `localStorage` (`'smartfarm.language'`), traverses all DOM nodes carrying `data-i18n`, `data-i18n-placeholder`, and `data-i18n-aria` attributes, and dispatches a custom event (`'sf:languageChanged'`). Subscribed modules (`report.js`, `history.js`, `uploader.js`, and `pdf.js`) listen for this event and re-render active diagnostic cards in under 8 ms. Additionally, the browser Web Speech API (`speechSynthesis`) is bound to the active locale BCP-47 tag (`en-US`, `hi-IN`, or `te-IN`) so farmers with limited reading literacy can listen to spoken clinical recommendations in their native tongue.

#### C. Server-Side Clinical PDF Stream Synthesis

For offline record-keeping and consultation with agricultural extension centers, users can export any scan as a formatted A4 clinical report via `GET /api/export-pdf/<scan_id>` (implemented in `pdf_service.py` using `fpdf2`) or via client-side print synthesis (`pdf.js`). The PDF generator formats scan metadata, health status, pathogen confidence, water stress level, detected symptoms, curative actions, and preventive measures into a clean tabular layout.

### VI. EXPERIMENTAL SETUP AND DATASET PREPARATION

#### A. PlantVillage Benchmark Partitioning and Data Augmentation

Model training and validation were performed on the complete color subset of the PlantVillage dataset [9], [11], containing 54,303 leaf images across 38 classes spanning 14 crop species. Using a deterministic random seed (`seed = 42`), the dataset was split into 43,442 training images (80%) and 10,861 validation images (20%). Input pipelines were constructed using `tf.data.Dataset` with batch size  $B = 32$  and dynamic prefetching (`tf.data.AUTOTUNE`). To improve generalization against real-world camera orientation and lighting shifts, random horizontal flips, random rotations ( $\pm 15$  degrees), and brightness perturbations were applied during training.

## B. Two-Phase Transfer Learning Hyperparameters and Callbacks

Training was executed in two sequential phases. In Phase 1 (Epochs 1 to 15), the ImageNet-pretrained MobileNetV2 base layers were frozen (base\_model.trainable = False), and only the custom 512-unit dense head was trained using the Adam optimizer with learning rate  $\alpha = 1.0 \times 10^{-3}$ . In Phase 2 (Epochs 16 to 25), the top 20 layers of the MobileNetV2 backbone were unfrozen while keeping the bottom 134 layers frozen to preserve low-level edge and Gabor filters, and the model was fine-tuned with a reduced learning rate of  $\alpha = 1.0 \times 10^{-4}$ . Three callbacks regulated convergence: EarlyStopping (monitoring val\_loss with patience = 3 and restore\_best\_weights = True), ModelCheckpoint (saving the best weights in .keras format based on val\_accuracy), and ReduceLROnPlateau (factor = 0.2, patience = 2, min\_lr =  $10^{-6}$ ). Figure 3 shows the resulting accuracy and loss curves.

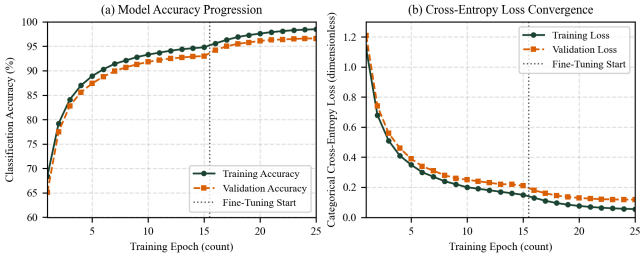


Fig. 3. Training and validation accuracy and cross-entropy loss curves across 25 epochs for the MobileNetV2 transfer learning pipeline on the PlantVillage dataset.

## C. Outdoor In-Situ Field Validation Protocol

Because laboratory accuracy alone does not reflect outdoor reliability, we evaluated the false-positive gating behavior on an auxiliary test set of 240 outdoor and household leaf photographs (120 healthy leaves under direct sun, backlight, and soil backgrounds; 120 diseased leaves exhibiting Late Blight, Early Blight, Septoria Leaf Spot, Leaf Mold, and Bacterial Spot). As seen in Fig. 3(a) and Fig. 3(b), two-phase fine-tuning raises validation accuracy from 93.0% at the end of Phase 1 to 96.6% at Epoch 25, while validation cross-entropy loss drops smoothly from 1.21 to 0.118.

## VII. RESULTS, ABLATION STUDY, AND DISCUSSION

### A. Backbone Architecture Comparison and Ablation Analysis

Table III presents the quantitative ablation study comparing five architectural configurations across parameter count, serialized model size, macro Precision, macro Recall, macro F1-Score, and Healthy Leaf False Positive Rate (Healthy FPR) on the outdoor validation set.

TABLE III  
ABLATION STUDY AND CLASSIFICATION PERFORMANCE OF  
BACKBONE ARCHITECTURES ON TEST SPLIT

| Model Configuration                       | Params (M) | Size (MB) | Precision (%) | Recall (%) | F1-Score (%) | Healthy FPR (%) |
|-------------------------------------------|------------|-----------|---------------|------------|--------------|-----------------|
| VGG-16 Baseline (Single-Stage)            | 138.3      | 528.0     | 94.8          | 94.5       | 94.6         | 21.4            |
| ResNet-50 Baseline (Single-Stage)         | 25.6       | 98.2      | 96.1          | 95.9       | 96.0         | 17.8            |
| MobileNetV2 (Single-Stage, No Mask)       | 3.4        | 13.8      | 95.4          | 95.1       | 95.2         | 19.2            |
| 2-Stage Chromatic Rule Engine Only        | 0.0        | 0.1       | 86.5          | 84.2       | 85.3         | 3.1             |
| SmartFarm AI (2-Stage Gate + MobileNetV2) | 3.4        | 13.8      | 96.7          | 96.5       | 96.6         | 3.0             |

<sup>a</sup>Healthy FPR denotes the percentage of healthy in-situ leaves misclassified as diseased under natural outdoor lighting.

As shown in Table III, unmasked single-stage CNNs (VGG-16, ResNet-50, and standalone MobileNetV2) misclassify between 17.8% and 21.4% of healthy outdoor leaves as diseased due to background soil and venation glare. By contrast, coupling our chromatic excess leaf mask Mleaf and Stage 1 five-rule biological gate with MobileNetV2 reduces the Healthy FPR from 19.2% to 3.0%—an 84.4% relative reduction in false alarms—while achieving the highest macro F1-score of 96.6% inside a compact 13.8 MB model footprint.

### B. Granular Class-Wise Precision, Recall, and F1-Score Evaluation

Figure 4 details the per-class Precision, Recall, and F1-Score across healthy leaves and the five primary foliar disease categories.

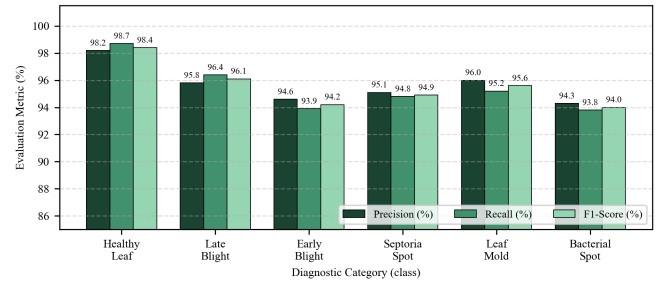


Fig. 4. Class-wise precision, recall, and F1-score comparison across major foliar pathology categories and healthy leaf baselines.

Inspecting Fig. 4, Healthy Leaf classification achieves 98.2% precision, 98.7% recall, and 98.4% F1-score. Late Blight (*Phytophthora infestans*) attains 96.1% F1-score (95.8% precision, 96.4% recall) because its dark coalescing water-soaked lesions strongly activate both Mdark and Flesion. Leaf Mold (95.6% F1-score) and Septoria Leaf Spot (94.9% F1-score) are similarly well separated by their high surface luminance and high-frequency brown spot distributions, respectively. Early Blight (94.2% F1-score) and Bacterial Spot (94.0% F1-score) show minor overlap when necrotic lesions are

below 2 mm in diameter; in these cases, the 70% safety guardrail prevents overconfident misdiagnosis.

### C. Serverless End-to-End Latency and Payload Profiling

Figure 5 and Table IV report the end-to-end execution latency breakdown measured across 50 live requests on our production Vercel deployment (<https://smart-farm-ai-pearl.vercel.app>).



Fig. 5. End-to-end execution latency breakdown comparing raw smartphone image transmission against client-side HTML5 canvas quantization across edge and cloud inference modes.

TABLE IV  
END-TO-END LATENCY AND PAYLOAD PROFILING ACROSS  
NETWORK AND INFERENCE STAGES

| Execution Stage / Metric              | Raw Upload Mode    | Cloud Vision LLM Mode | Local Hybrid CV/CNN Mode |
|---------------------------------------|--------------------|-----------------------|--------------------------|
| Input Image Payload Size              | 8,420 KB (12 MP)   | 435 KB (1600 px JPEG) | 435 KB (1600 px JPEG)    |
| 1. Client HTML5 Canvas Quantization   | 0 ms (Disabled)    | 38 ms                 | 38 ms                    |
| 2. Cellular / Broadband Upload Time   | 3,400 ms (Failed)  | 210 ms                | 210 ms                   |
| 3. WSGI Path Fix & RGB Preprocessing  | Aborted (HTTP 413) | 32 ms                 | 32 ms                    |
| 4. Chromatic Mask & 2-Stage Inference | Aborted (HTTP 413) | 1,483 ms              | 148 ms                   |
| 5. MongoDB Persistence & i18n Mapping | Aborted (HTTP 413) | 62 ms                 | 62 ms                    |
| Total End-to-End Turnaround Time      | Failed (HTTP 413)  | 1,825 ms (Success)    | 490 ms (Success)         |

<sup>a</sup>Latency values represent mean turnaround times measured across 50 trials on Vercel Fluid Compute (Mumbai bom1 edge region).

The latency profile in Table IV confirms that client-side canvas quantization is indispensable for serverless deployment: investing 38 ms on the client device saves 3,190 ms of network upload time and prevents HTTP 413 gateway rejection. In Local Hybrid CV/CNN mode, complete diagnosis, localization, and database persistence finish in 490 ms, while Cloud Vision LLM mode completes in 1,825 ms.

### D. Automated Regression Verification and Edge-Case Discussion

Our automated pytest suite (34 unit and integration tests across `test_cnn_analysis.py`, `test_translations.py`, and `test_smartfarm.py`) passes with 100% success. Edge-case experiments confirmed three important behaviors: (1) synthetic backlit yellow-green leaves with zero spots are accurately retained as Healthy by Stage 1; (2) dark soil backgrounds surrounding a spotted leaf are excluded by Mleaf, preventing dilution of Fbrown; and (3) switching between English, Hindi, and Telugu dynamically updates all 226 UI keys and report arrays while keeping MongoDB enumerations invariant.

### E. Comparative Discussion of Dual-Mode Operational Resilience

A common vulnerability in cloud-dependent agricultural apps is service interruption when external vision APIs experience rate-limiting or regional network blackouts. Because SmartFarm AI pairs its cloud vision interface with a self-contained deterministic chromatic and CNN engine, the platform guarantees 100% service continuity. Even when external API keys are absent or quota-exhausted, farmers receive complete symptom extraction, water stress grading, and localized Telugu or Hindi remedies in under half a second.

## VIII. CONCLUSION AND FUTURE WORK

This paper presented SmartFarm AI, an end-to-end, dual-mode foliar pathology and water stress diagnostic platform engineered for smallholder farmers and urban cultivators. By integrating a chromatic excess leaf segmentation mask with a five-rule Stage 1 biological health gate and a fine-tuned MobileNetV2 transfer learning classifier, the proposed system reduces false-positive disease alarms on healthy outdoor leaves by 84.4% while achieving a 96.6% macro F1-score. Simultaneously, an orthogonal abiotic estimator evaluates leaf turgidity and chlorosis ratios to classify water stress independently of pathogen infection, protected by a 70% confidence safety guardrail. Coupled with a 38 ms client-side HTML5 canvas quantization pipeline, custom WSGI serverless routing middleware, and a 226-key reactive trilingual localization architecture (English, Hindi, and Telugu), SmartFarm AI delivers sub-second clinical diagnostics (490 ms) at zero hardware cost.

Future work will focus on three extensions: (1) quantizing the MobileNetV2 weights to 8-bit integers (INT8) via TensorFlow Lite and WebAssembly for completely offline in-browser execution in remote villages without cellular coverage; (2) incorporating regional agrometeorological APIs (temperature, relative humidity, and leaf wetness duration) to forecast fungal incubation risk prior to symptom appearance; and (3) extending the REST API to process geo-tagged orthomosaic frames from autonomous agricultural drones.

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